

Experimental Protocol

LoRaWAN Signal Strength Test in Subsurface Conditions

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Objective of the Experiment

In the context of the planned deployment of a sensor module intended to be installed at a depth of approximately 5 cm below the earth's surface, the central question arises as to the extent to which partial burial of the system affects the quality of wireless communication via the LoRaWAN protocol. In practical applications, subterranean installation serves to protect the sensor from external influences such as vandalism, weather conditions, or mechanical damage caused by maintenance activities in the surrounding area—such as lawn mowing or the use of root tools during tree maintenance.

The aim of the present experiment is therefore to systematically investigate the influence of embedding a LoRaWAN-based sensor system at various depths beneath the ground surface on signal transmission. The primary focus lies on determining whether reliable operation can be ensured at typical installation depths or whether significant signal degradation is to be expected.

Two different types of antennas are compared: a standard rod antenna that comes bundled with the microcontroller used, and an alternative, longer antenna that is claimed to offer higher reception sensitivity. By directly comparing both variants under identical test conditions, the study seeks to determine the extent to which alternative antenna designs can positively impact signal quality in subterranean operation.

Experimental Setup

To create a controlled and reproducible environment, the experimental setup is designed to allow all investigations to be conducted under laboratory conditions. The experiments are to be carried out in a plastic container with a volume of 12 liters, which will be filled with soil. To replicate the region-specific soil conditions as realistically as possible, it is intended to use earth sourced directly from the campus grounds of Flensburg University of Applied Sciences. This soil naturally consists of a mixture of sand, stones, and humus. After filling, the soil will be compacted uniformly using body weight of approximately 85 kg to simulate the typical compaction that occurs during backfilling processes.

Before the actual antenna tests begin, the volumetric water content of the soil will be measured using an SMT100 sensor in order to establish a reference value for the natural moisture content. No signal tests are planned during this initial phase.

The first test series will begin with a defined moistening of the soil. For this purpose, the maximum documented rainfall in the Flensburg area—approximately 40 liters per square meter—will be scaled to the surface area of the container [1] [2]. This results in a

calculated water volume of 3.2 liters to be evenly distributed across the soil. This scenario represents a modeled moisture condition following an extreme rainfall event.

In a subsequent phase, the amount of water will be incrementally increased until the soil reaches an almost fully saturated, homogeneous, and viscous consistency. This phase aims to investigate the signal behavior under conditions of extreme soil saturation.

Before each measurement, a reference signal test will be conducted in open air using both antenna types in order to document the maximum possible signal quality without soil interference. Subsequently, one antenna at a time will be inserted into the center of the container at a depth of 150 mm. After a defined period, the antenna will be raised in 50 mm increments, allowing for measurements at burial depths of 150 mm, 100 mm, and 50 mm.

At each test depth, the tip of the antenna is surrounded horizontally by approximately 157 mm of soil, while vertically it is covered by 150 mm, 100 mm, or 50 mm of earth, respectively. Beneath the container is a solid reinforced concrete slab, which serves as an effective barrier against downward signal emission. This ensures that the signal must propagate entirely through the surrounding soil, allowing for an isolated analysis of the attenuation properties of the earth material.

Experimental Procedure

At the beginning of the experiment, a commercially available plastic container with a capacity of 12 liters was completely filled with soil. Subsequently, the volumetric water content of the soil sample was determined using an SMT100 sensor. The measurement yielded an initial moisture content of approximately 2 vol%, which classifies the soil as dry.

In the next step, the previously calculated water volume of 3.2 liters was evenly distributed over the soil surface to simulate a moisture level typically observed after a heavy rainfall event. Once the water had fully infiltrated the soil, a second measurement was conducted using the SMT100 sensor. This yielded an average volumetric soil moisture of 19.8 %.



Abbildung 1:SMT100

After the water application, the soil was once again compacted by applying a body weight of approximately 85 kg in order to realistically simulate the typical consistency of backfilled soil in buried installations.

For the signal strength measurements, an Heltc ESP32 microcontroller equipped with a LoRaWAN module was employed. The device was alternately fitted with each of the two antenna types under investigation and programmed with the necessary firmware for wireless data transmission. Initially, the microcontroller was positioned such that the antenna was located at the rim of the container but above the soil surface. In this reference position, the first set of measurements was conducted to capture the undamped signal strength under optimal conditions.

Subsequently, the antenna was inserted into the center of the container to a depth of 150 mm, so that it was completely surrounded by soil. After a defined period, the antenna was gradually raised in 50 mm increments, with further measurements conducted at depths of 100 mm and 50 mm. This procedure was repeated identically for both antenna types to ensure the comparability of measurement results.



Abbildung 3: long antenna



Abbildung 2: short antenna

After completion of the first series of tests, the soil was intentionally saturated with additional water until the SMT100 sensor registered a maximum soil moisture content of 46.68 %. Under these conditions—which correspond to an almost fully saturated soil environment—all measurements were repeated in the same manner.



Abbildung 4: SMT100 sensor in soil with 46% volumetric water content

Results

The measurement results clearly demonstrate that both soil moisture content and the antenna burial depth exert a significant influence on the quality of signal transmission. RSSI values were recorded for four distinct antenna positions: in open air (serving as a reference), and at burial depths of 150 mm, 100 mm, and 50 mm. The results were obtained under two defined soil moisture conditions: 19.6 % (moist) and 46 % (highly water-saturated). Furthermore, a distinction was made between two types of antennas—one longer variant and a shorter standard antenna.

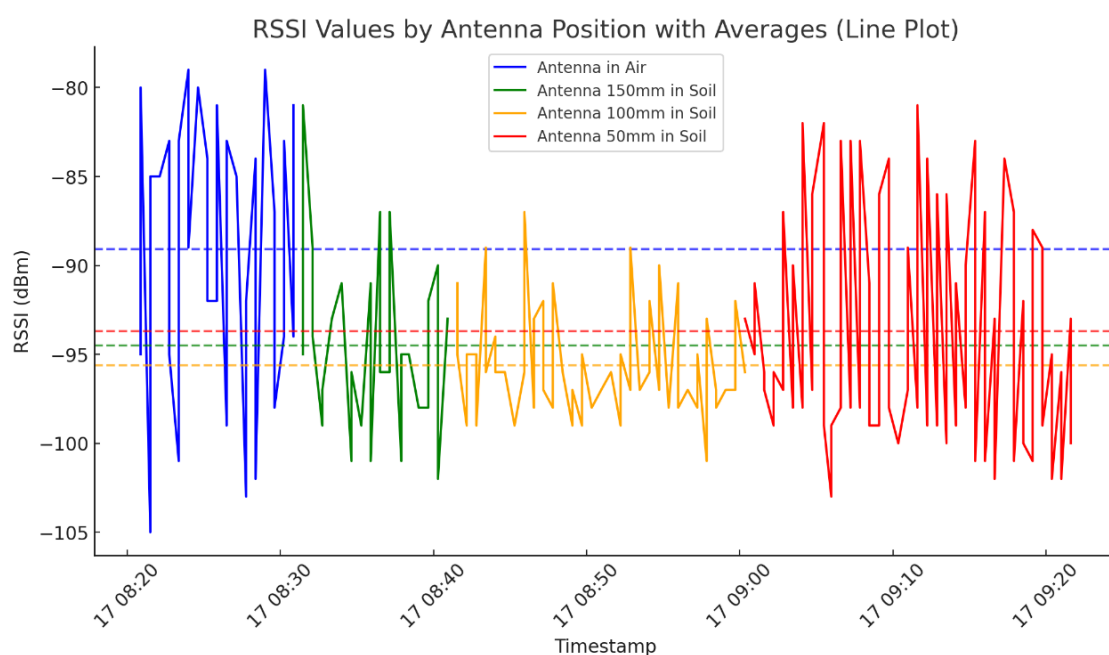


Abbildung 5: Long antenna in soil with 19% volumetric water content

For the longer antenna, a pronounced attenuation of signal strength as a function of burial depth was observed at a soil moisture content of 19.6 %. While the mean RSSI value measured in open air was approximately -89 dBm, it deteriorated to -95.6 dBm at a burial depth of 150 mm. At a depth of 100 mm, the average RSSI was -95.3 dBm, and at 50 mm, a value of -94.4 dBm was recorded.

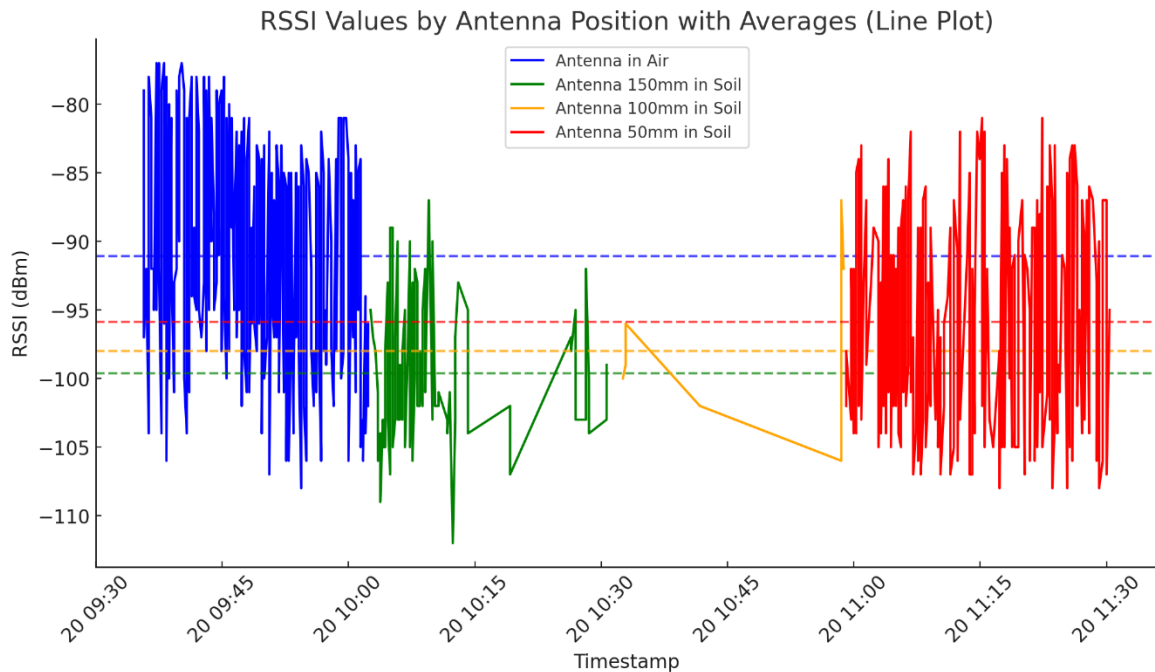


Abbildung 6: Long antenna in soil with 46% volumetric water content

The differences observed indicate that signal attenuation increases with greater burial depth, although not in a linear fashion. Even at moderate moisture levels, a clear influence of the surrounding soil on signal transmission can be identified. Nevertheless, the longer antenna demonstrated comparatively stable transmission performance across all positions.

The situation becomes significantly more critical under conditions of high soil saturation at a moisture content of 46 %. In this case, even the reference measurement in open air deteriorated to -94.4 dBm, which may be attributed to environmental noise or minor electromagnetic interference. The RSSI values measured in soil were substantially lower: at a depth of 150 mm, an average of -105.1 dBm was recorded, while measurements at 100 mm and 50 mm yielded -103.9 dBm and -102.8 dBm, respectively. The maximum deviation from the air measurement thus exceeded 15 dB. These findings clearly indicate that signal attenuation increases significantly under conditions of high soil moisture. Nonetheless, the longer antenna still produced relatively robust results under such adverse conditions.

In contrast, the shorter antenna exhibited a considerably more sensitive response to the test conditions. At a soil moisture level of 19.6 %, the mean reference RSSI in open air was -96.3 dBm. The measurements obtained in soil were notably lower: at 150 mm

depth, an average of -105.1 dBm was observed; at 100 mm, -100.6 dBm; and at 50 mm, -99.7 dBm. These values clearly demonstrate that the shorter antenna delivers significantly reduced reception performance at comparable depths in comparison to the longer variant.

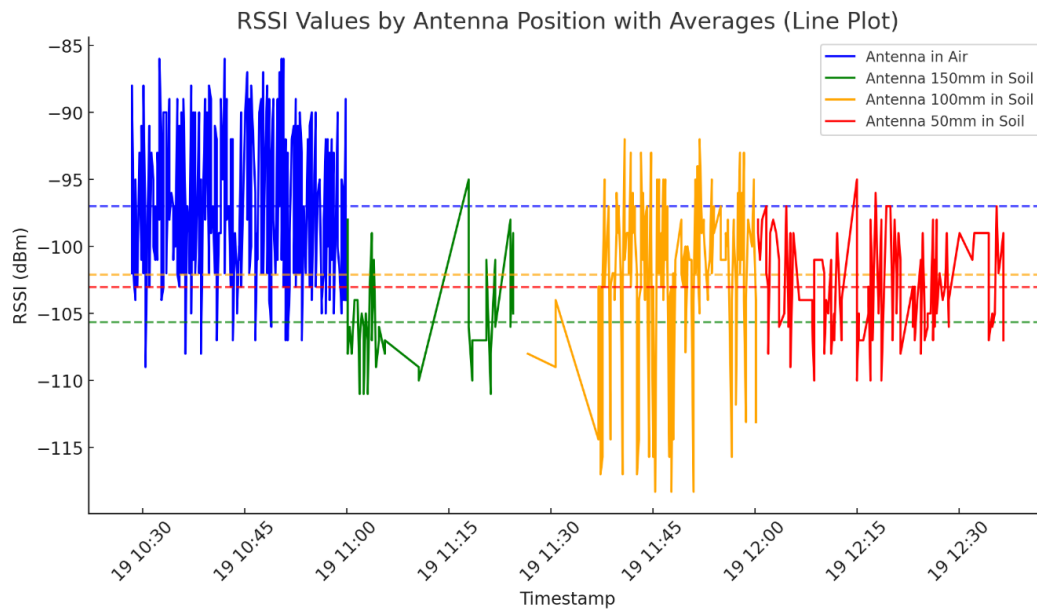


Abbildung 7: Short antenna in soil with 19% volumetric water content

The effect was even more pronounced at a soil moisture content of 46 %. The reference measurement in open air showed a mean RSSI value of -97.2 dBm. The values recorded in soil were -106.2 dBm at 150 mm burial depth, -105.4 dBm at 100 mm, and -104.5 dBm at 50 mm. These results demonstrate that the short antenna is barely capable of transmitting a sufficiently strong signal under conditions of high soil moisture. The attenuation is markedly greater compared to the long antenna and suggests a lack of sufficient performance margin.

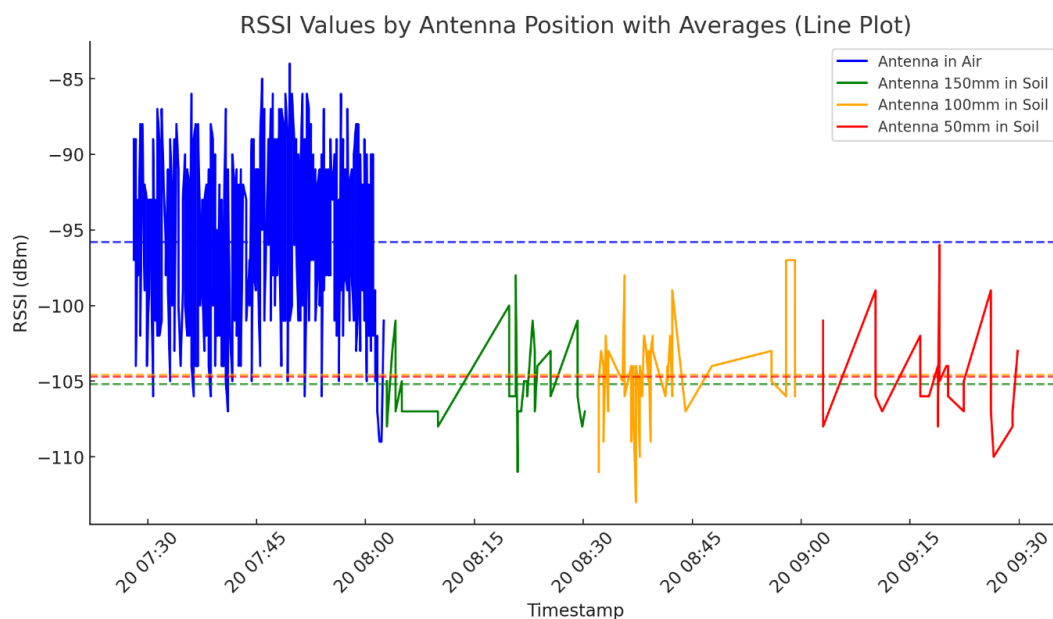


Abbildung 8: Short antenna in soil with 46% volumetric water content

Conclusion

The conducted measurement series clearly demonstrate that both the burial depth and soil moisture content have a significant impact on the signal strength of a LoRaWAN transmitter operating underground. Nevertheless, based on the data obtained, it can be concluded that operation under realistic conditions—particularly at a burial depth of 50 mm—is fundamentally feasible and functional.

The comparison between the two antenna types revealed that the longer antenna consistently exhibited significantly greater reception stability across all scenarios. Even in highly water-saturated soil, it was capable of maintaining adequate signal transmission. In contrast, the shorter standard antenna proved to be substantially more susceptible to performance degradation in deep and moist soil layers and should therefore not be preferred for such applications.

Particularly relevant for practical implementation is the finding that data transmission was achievable with both antenna types at a burial depth of 50 mm, although the longer antenna consistently yielded superior results. The measurements confirm that, under these conditions, stable operation is possible and that the communication link can be maintained even at elevated soil moisture levels.

Although the RSSI values measured at a depth of 50 mm show a certain degree of attenuation compared to the open-air reference, they remain within the range of practical applicability. Even under adverse environmental conditions, the communication link was not interrupted, indicating sufficient signal integrity.

Overall, it can thus be concluded that the intended installation depth of 50 mm represents a viable compromise between mechanical protection and signal accessibility. The central objective of the experiment—evaluating transmission quality under partial burial of the sensor—was successfully addressed. Taking into account the defined boundary conditions and assuming the use of a high-performance antenna, there are no fundamental objections to the underground installation of the system at a depth of 50 mm.

References

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